engineering 4 us all

Student Handout

Name: Anthony Claudet **Date:** _____ **Class:** _____

MyDesign Portfolio

Element D: Design Concept Generation, Analysis, and Selection

Design Idea Generation *(D1)*

Design Idea #1

Date: 4/24/25
Description: **A simple program that collects a list of devices with their names, power usage (kW), and essentiality. Then it checks the time of day (peak or not) and outputs whether each device should be delayed or kept on.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:

[user input]
⬇
[device list created]
⬇
[peak usage checker]
⬇
[recommendation output]

Scale: one screenfull of text

Materials: Matlab programming, home systems equipment

Dimensions: ~30-50 lines of code

Design Idea #2

Date: 4/24/25
Description: **A basic GUI that shows a dashboard of devices. Each device appears as a colored box, where green = stay on, yellow = delay suggested, red = turn off. The user can click on devices to see more info.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:

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[Device 1]

[Device 2]

[Device 3]

Scale: 3x2 window screen

Materials: Matlab programming

Dimensions: GUI layout

Design Idea #3

Date: 4/25/25
Description: **A program that calculates total energy saved (in kWh) if all delay recommendations are followed during peak hours.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:

[device list]
⬇
[delay vs stay decision]
⬇
[sum of energy saved]
⬇
[display savings]

Scale: basic math functions

Materials: Matlab programming

Dimensions: ~30-50 lines of code

Design Idea #4

Date: 4/27/25
Description: **Program allows users to edit the properties of a device after it's been entered. Useful if a user makes a mistake during input.**
Drawing that includes distinguishing information such as scale, dimensions, and materials:

[device list]
⬇
[edit button for device]
⬇
[change properties]

Scale: interactive console prompts

Materials: Matlab programming

Dimensions: ~30-50 lines of code

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Design Idea Comparison and Selection *(D1, D2)*

	weight	Design Idea 1	Design Idea 2	Design Idea 3	Design Idea 4
Requirement A (req #1) - the program must evaluate each device and make recommendations	X	Significant step toward req #1	Small factor of req #1	Calculation factor of req #1	External factor. Does not meet req #1
Requirement B (req #2) - the program must allow user input to define device properties	X	Perfectly satisfies req #2, prompts user input	External factor. Does not meet req #2.	External factor. Does not meet req #2	Significant factor of req #2. Allows edits
Requirement C (req #3) - the program must categorize time of day as either "peak" or "off-peak" hours based on user input and adjust recommendations accordingly.	X	Small factor of req #3, only checks if peak or off-peak	Significant factor of req #3. Box colors are dependent on peak or off-peak values	Significant factor of req #3. Calculates energy saved by using peak or off-peak values	Irrelevant to req #3
Requirement D (req #4) - the program	X	Perfectly satisfies req	Perfectly satisfies req	Small factor of req #4,	Irrelevant to req #4

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must output clear recommendations for each device.		#4, outputs suggestions	#4, where colors show suggestions	performs calculations	
Requirement E (req #5) - The program must be able to analyze at least five devices in one session without crashing or freezing.	X	If the program can evaluate five devices or more, req #5 is satisfied	If it can display five or more boxes, req #5 is satisfied	Irrelevant to req #5	Small factor of req #5
Score out of 10 (each of the five requirements is worth 2 pts. If it perfectly satisfies or is highly relevant to a req, it gets +2 pts. If it is a small factor of the req, it gets +1 pt. If it is irrelevant, +0 pts.	X	9/10	7/10	5/10	3/10

Our team chooses design #1 to solve the community problem because it fulfills all requirements except for a small part of requirement #3, in which it should categorize time of day as peak or off-peak.

Design Blueprint *(D4)*

The prototype I selected combines the strengths of design ideas #1 and #3 to create a user-friendly, text-based smart energy optimizer that fulfills all my requirements. The program will allow users to input device names, power usage (in kW), and whether each is essential or non-essential. It will automatically determine whether it is peak or off-peak time, removing the

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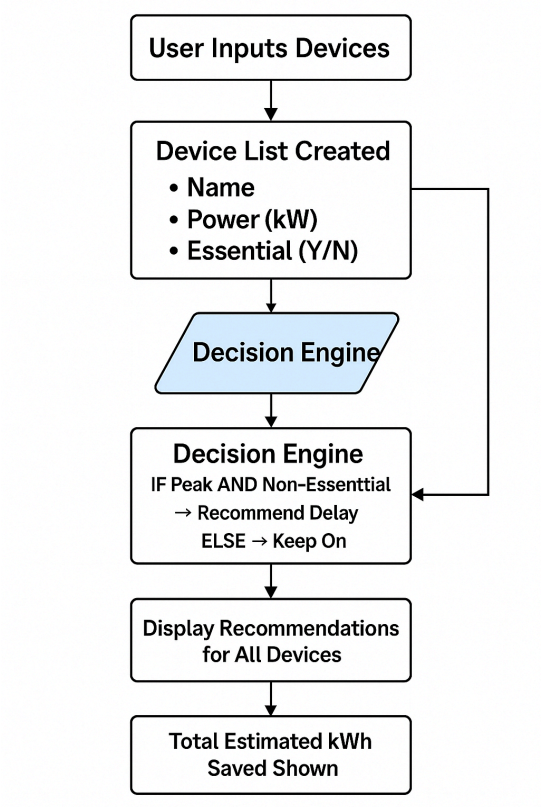
need for manual input. Based on this, the program will recommend whether to delay, keep on, or turn off each device, clearly displaying the result. Additionally, it will calculate the total potential energy savings (in kWh) if all delay recommendations are followed, giving users measurable feedback. This prototype satisfies the high-priority design requirements including accurate recommendations, flexible input, and clear output, while introducing a useful bonus feature (energy savings calculation) from a low-priority requirement.

Drawing: